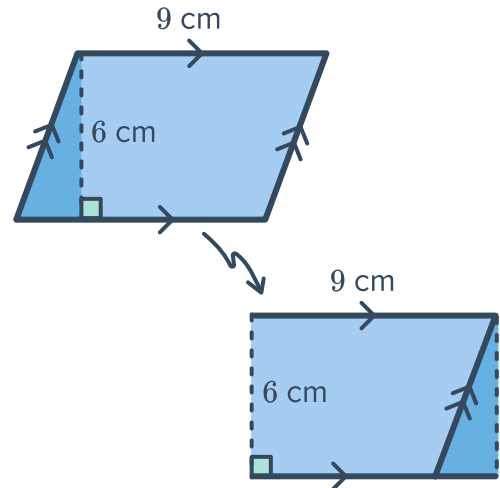


Parallelograms



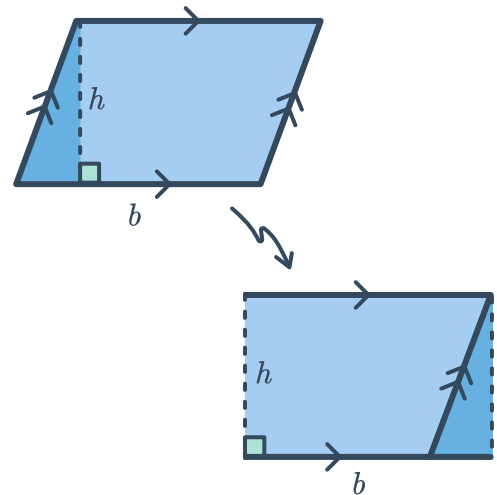
- 1 The given parallelogram is formed into a rectangle:

- a Find the area of the rectangle.
- b Hence, find the area of the parallelogram.



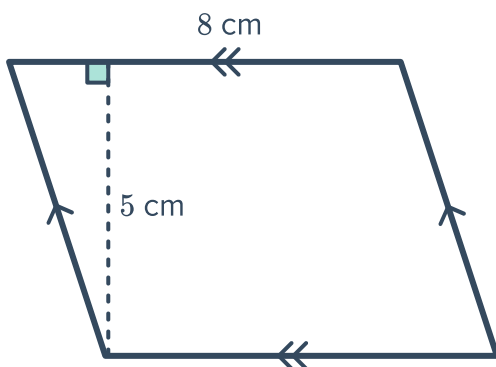
- 2 The given parallelogram is formed into a rectangle:

- a Find an expression for the area of the rectangle in terms of b and h .
- b Hence, find an expression for the area of the parallelogram in terms of b and h .

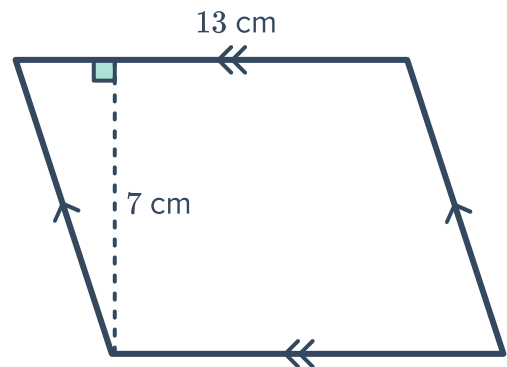


- 3 Find the area of the following parallelograms:

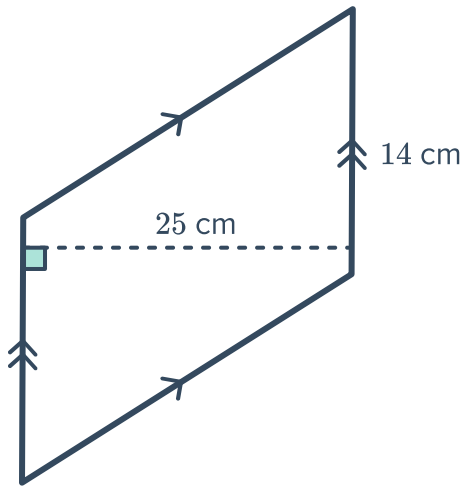
a



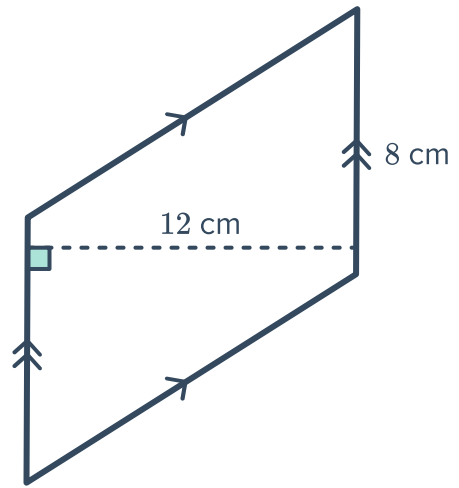
b



c



d



4 Determine whether the following pairs of values could be the dimensions of a parallelogram with an area of 70 mm^2 .

a Base = 10 mm, Height = 7 mm

b Base = 7 mm, Height = 10 mm

c Base = 1 mm, Height = 70 mm

d Base = 2 mm, Height = 70 mm

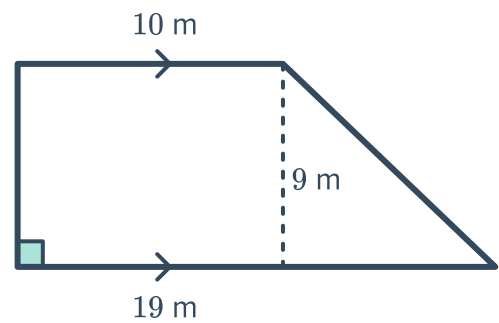
Trapeziums

5 The given trapezium is split into a rectangle and a right-angled triangle:

a Find the area of the rectangle.

b Find the area of the triangle.

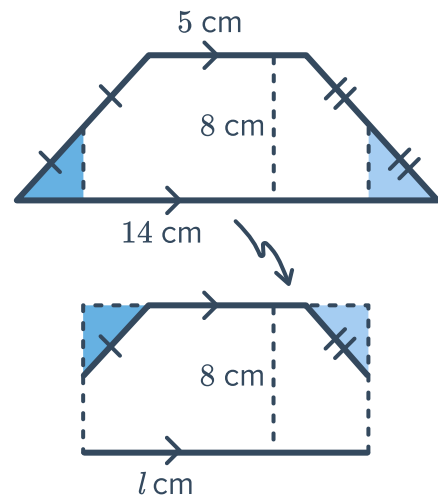
c Hence, find the area of the trapezium.



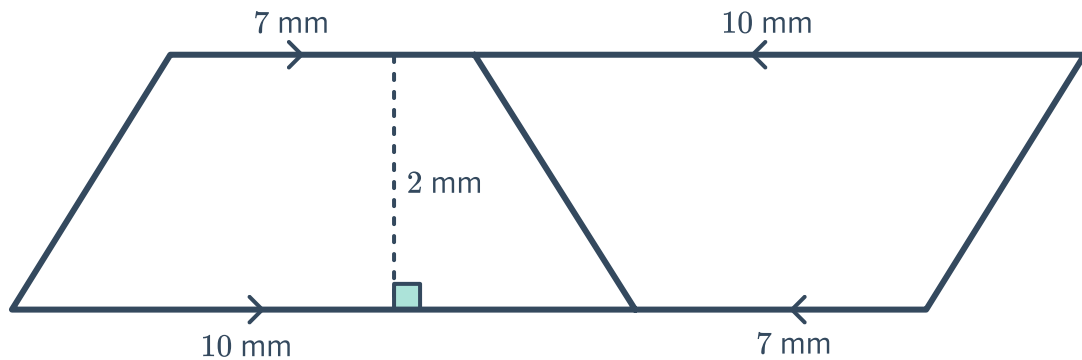
- 6 The given trapezium is formed into a rectangle:



- Find the length, l , of the rectangle.
- Hence, find the area of the trapezium.

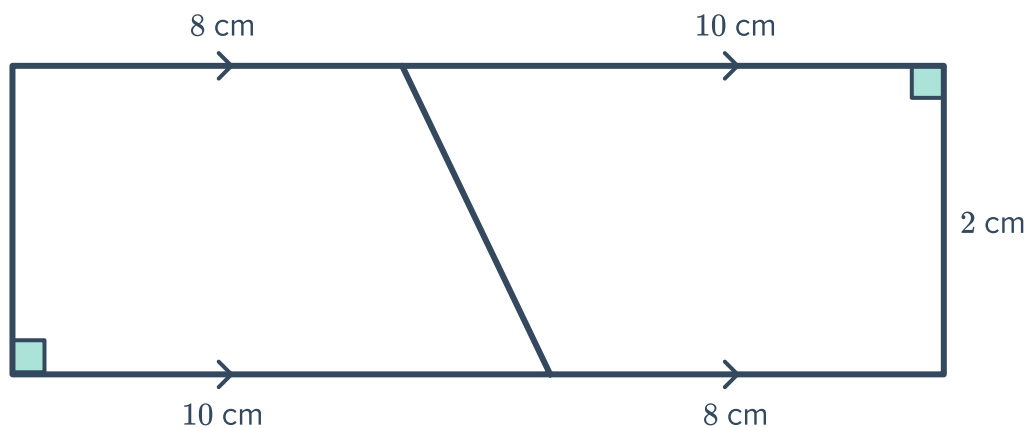


- 7 Two identical trapezia are put together to make a parallelogram:



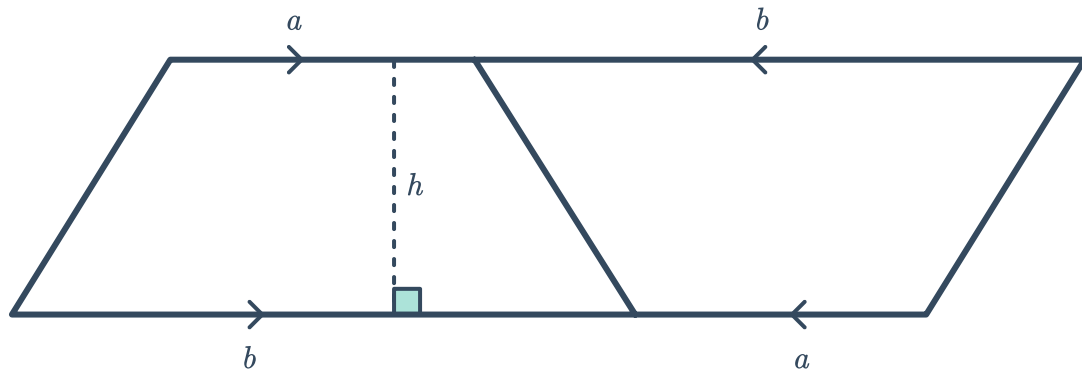
- Find the area of the entire parallelogram.
- Find the area of one of the trapezia.

- 8 Two identical trapezia are put together to make a rectangle:



- Find the area of the entire rectangle.
- Find the area of one of the trapezia.

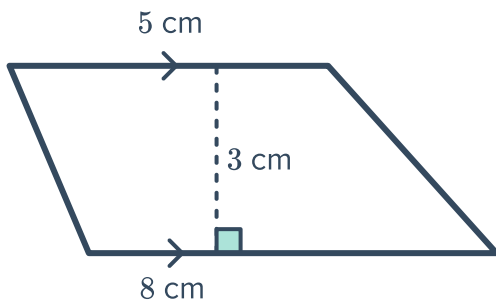
9 Two identical trapezia are put together to  a parallelogram:



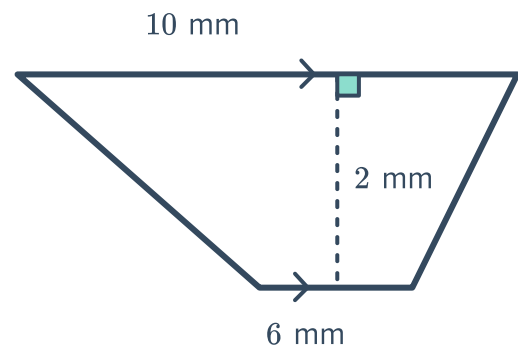
- a Find an expression for the area of the entire parallelogram in terms of a, b and h .
- b Find an expression for the area of one trapezia in terms of a, b and h .

10 Find the area of the following trapeziums:

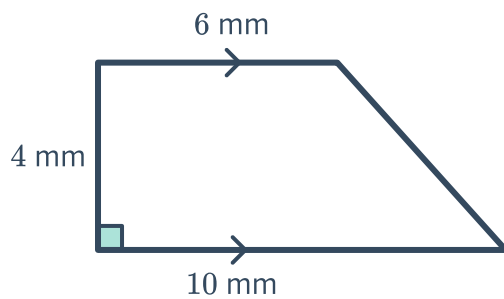
a



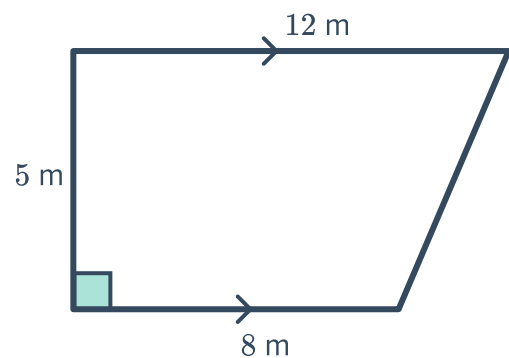
b



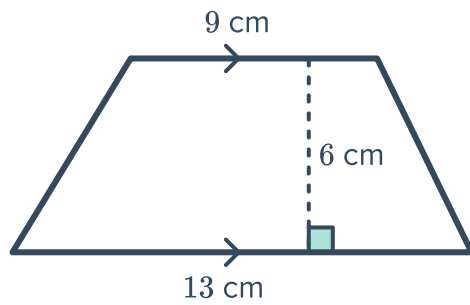
c



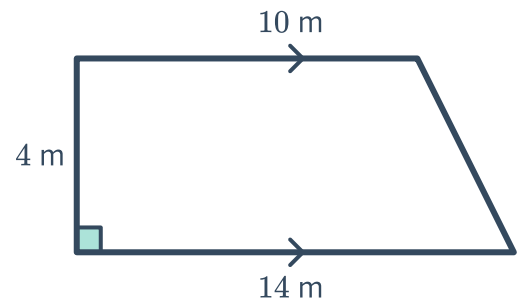
d



e



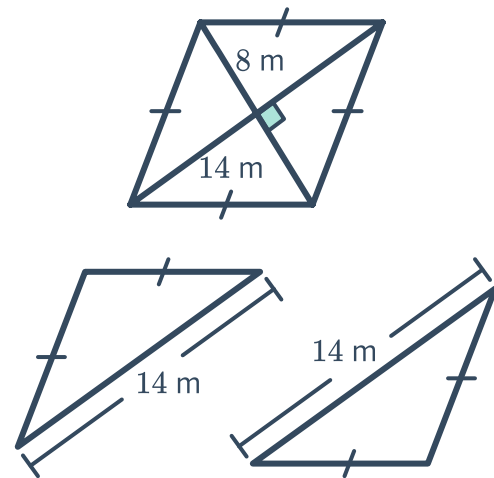
f



Rhombuses

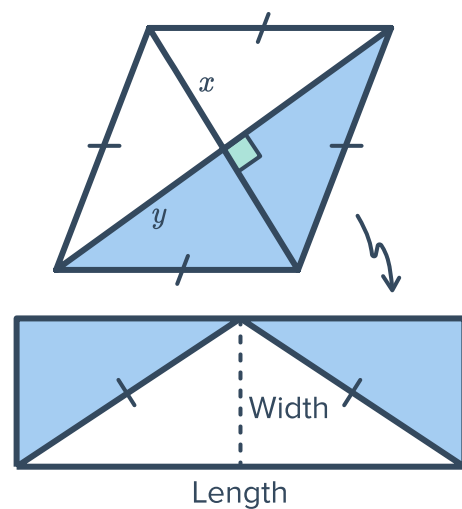
- 11 The given rhombus can be split into two triangles:

- Find the area of one triangle.
- Hence, find the area of the rhombus.



- 12 The given rhombus is formed into a rectangle:

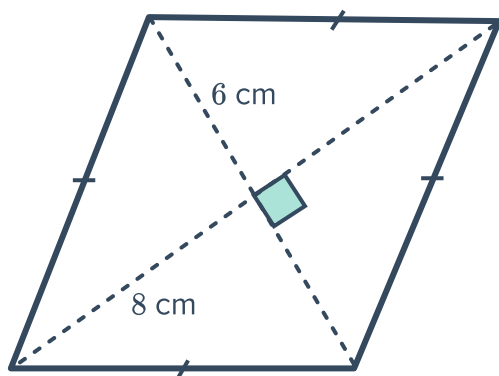
- Find the length of the rectangle in terms of y .
- Find the width of the rectangle in terms of x .
- Find the area of the rectangle in terms of x and y .
- Hence, find the area of the rhombus in terms of x and y .



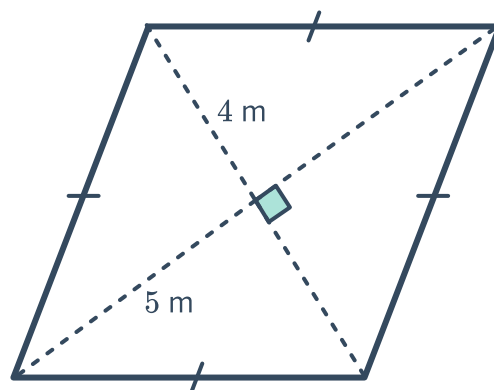
13 Find the area of the following rhombuses:



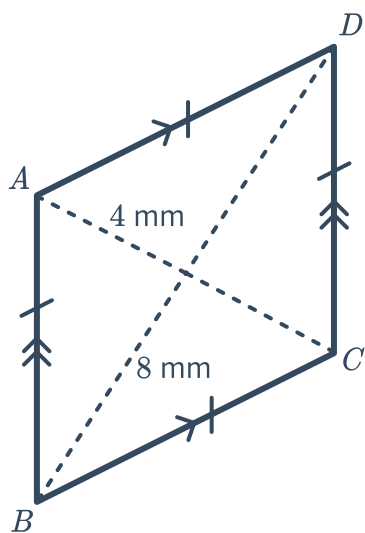
a



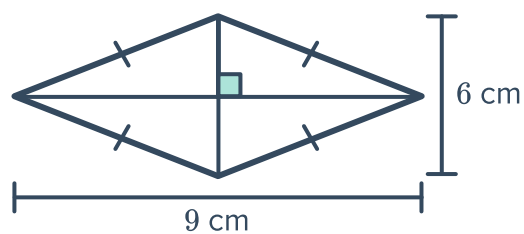
b



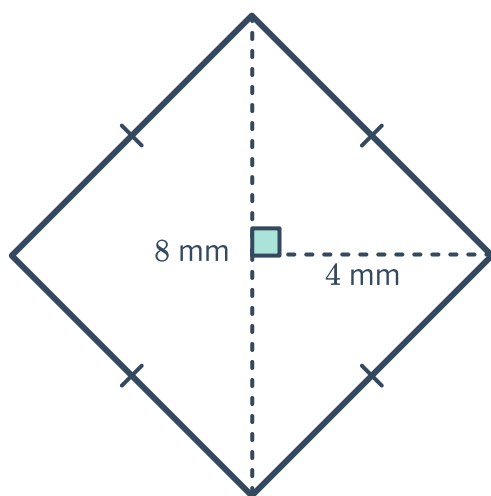
c



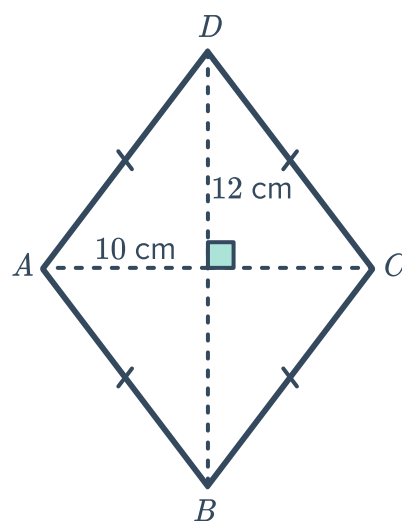
d



e



f



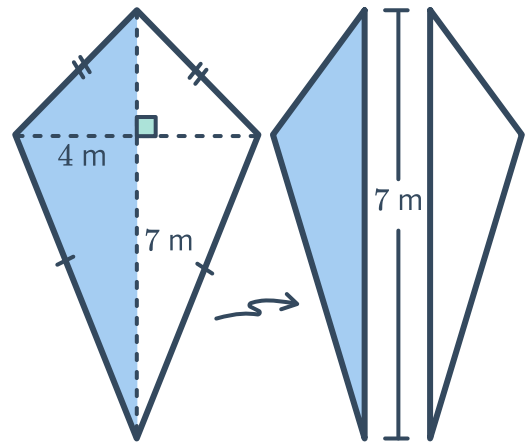
- 14 Determine whether the following pairs of s could be the diagonal lengths, x and y of a rhombus with an area of 9 m^2 .

- a $x = 2\text{ m}$ and $y = 9\text{ m}$.
b $x = 6\text{ m}$ and $y = 3\text{ m}$.
c $x = 12\text{ m}$ and $y = 3\text{ m}$.
d $x = 6\text{ m}$ and $y = 6\text{ m}$.

Kites

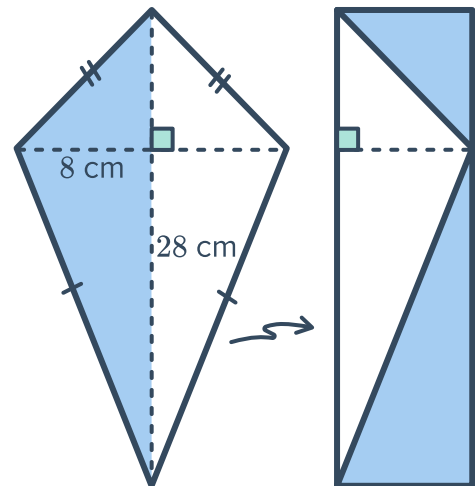
- 15 The given kite can be split into two triangles:

- a Find the area of one of the triangles.
b Hence, find the area of the kite.



- 16 The given kite is formed into a rectangle:

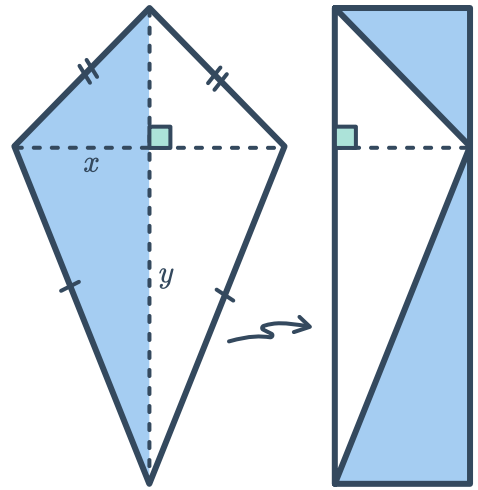
- a Find the length of the rectangle.
b Find the width of the rectangle.
c Hence, find the area of the kite.



17 The given kite is formed into a rectangle:

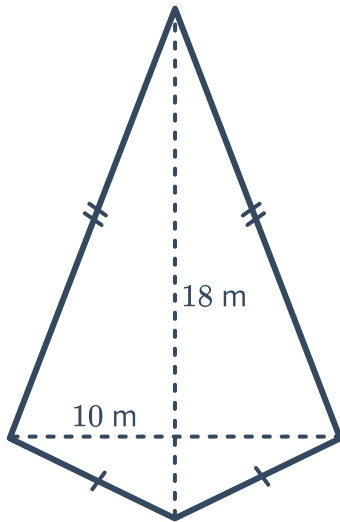


- Find the length of the rectangle in terms of y .
- Find the width of the rectangle in terms of x .
- Find the area of the rectangle in terms of x and y .
- Hence, find the area of the kite in terms of x and y .

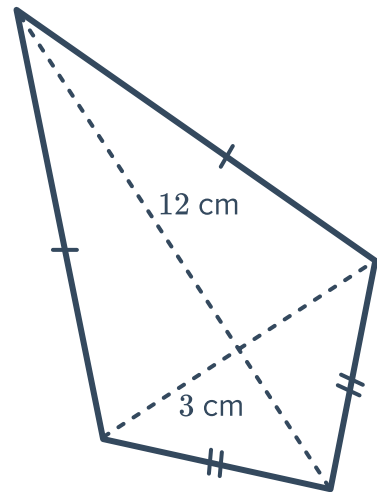


18 Find the area of the following kites:

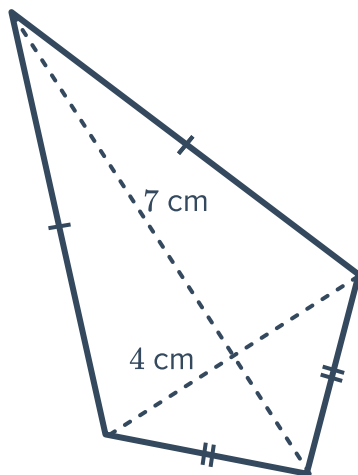
a



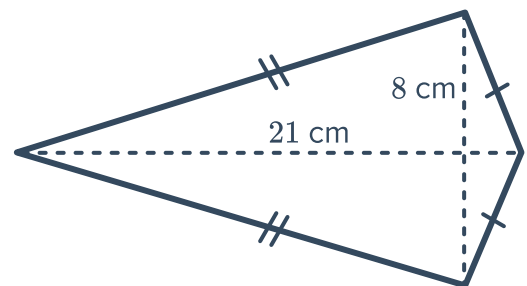
b



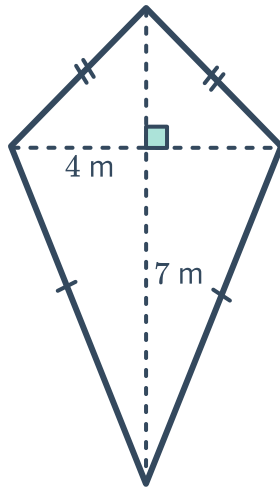
c



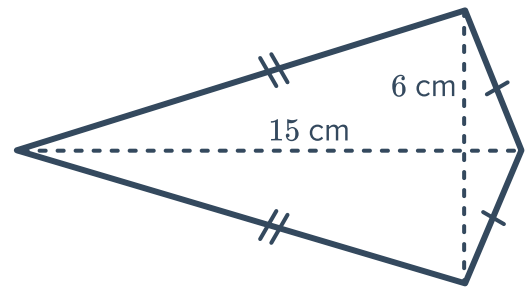
d



e



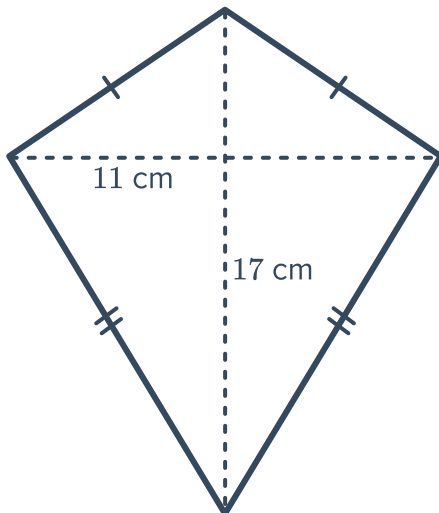
f



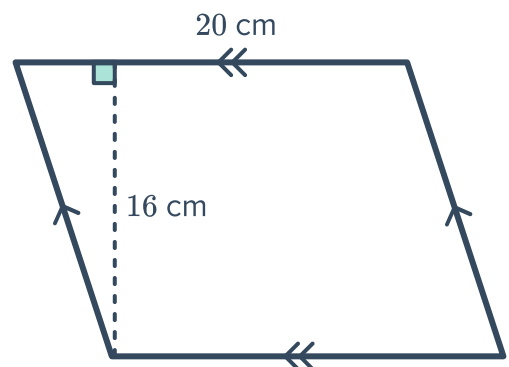
Mixed areas

19 Find the area of the following quadrilaterals:

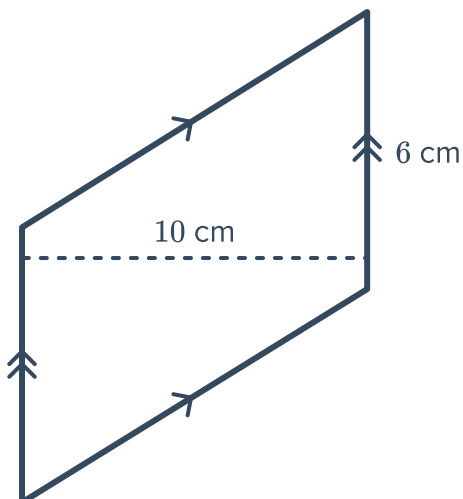
a



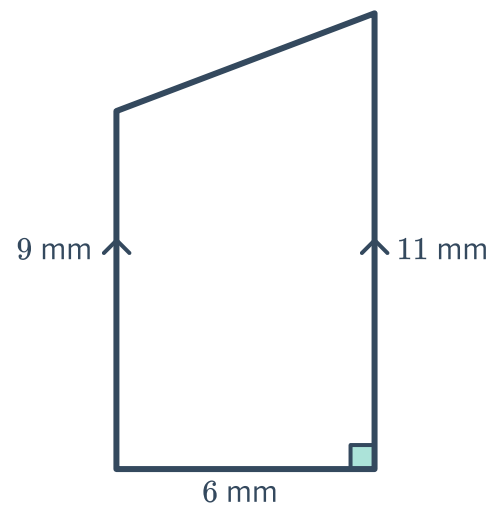
b



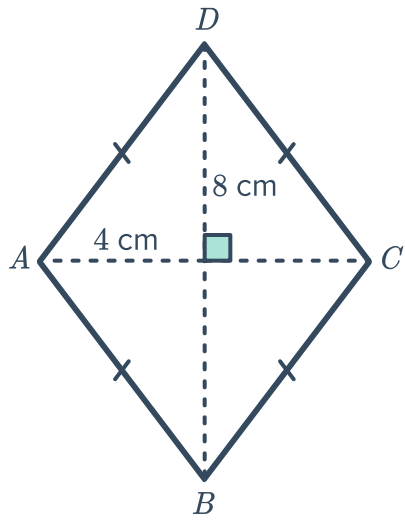
c



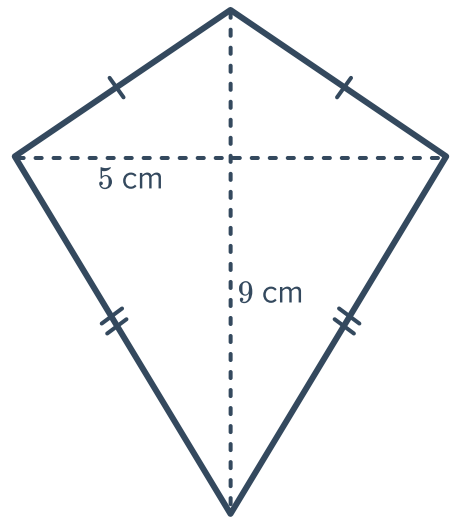
d



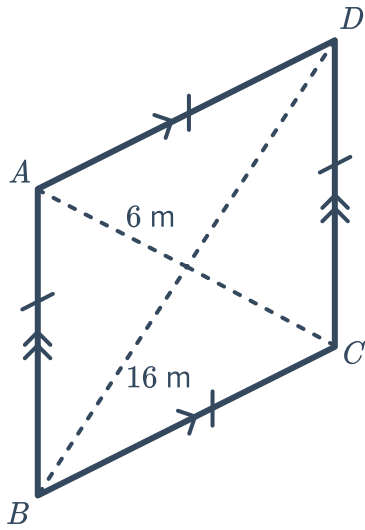
e



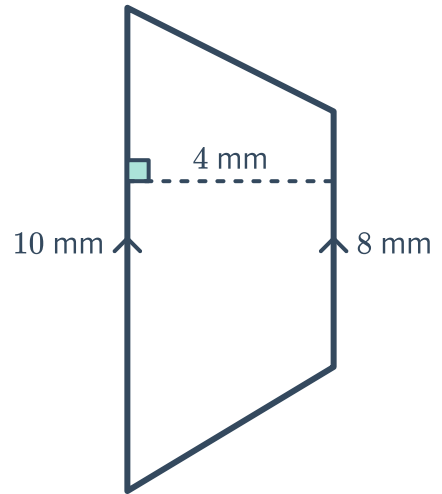
f



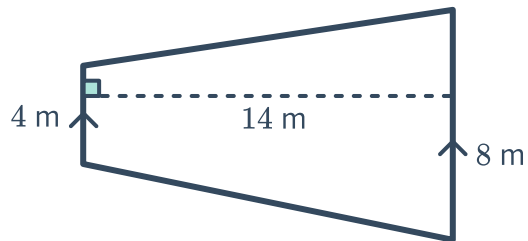
g



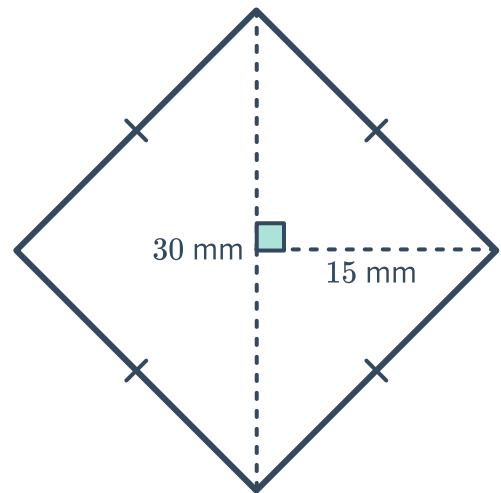
h



i



j

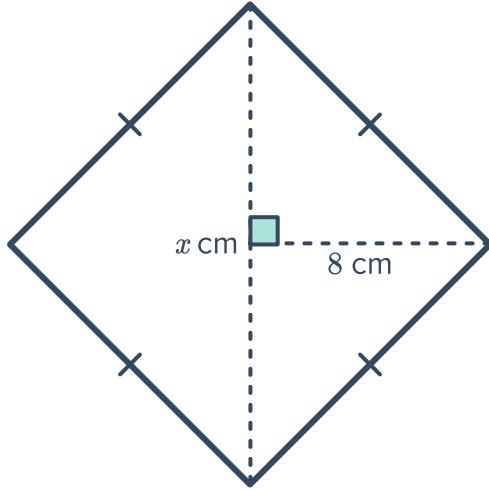




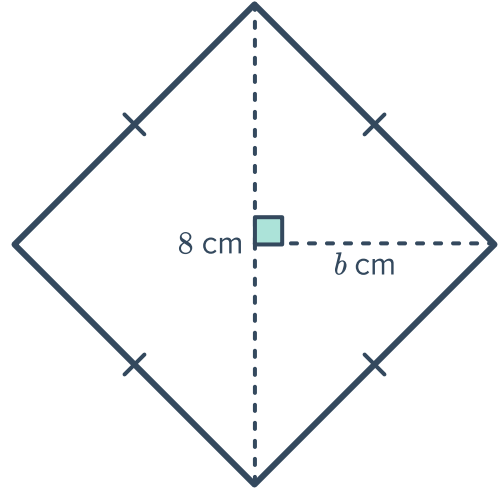
Missing lengths

20 For each of the following rhombuses, find the value of the pronumeral:

a $A = 64 \text{ cm}^2$

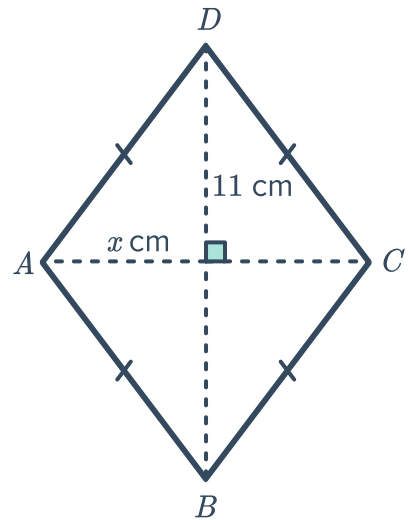


b $A = 128 \text{ cm}^2$



21 Rhombus $ABCD$ has an area of $A = 55 \text{ cm}^2$:

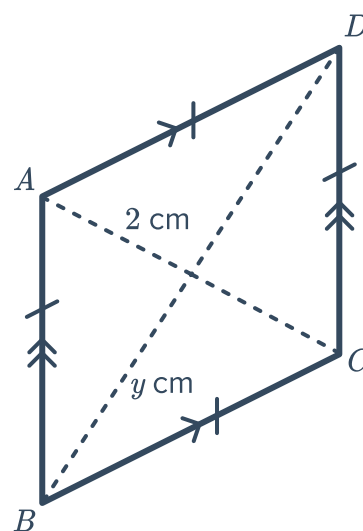
Given the diagonal $BD = 11 \text{ cm}$, and $AC = x \text{ cm}$, find the value of x .



22 Rhombus $ABCD$ has an area of 13 cm^2 :

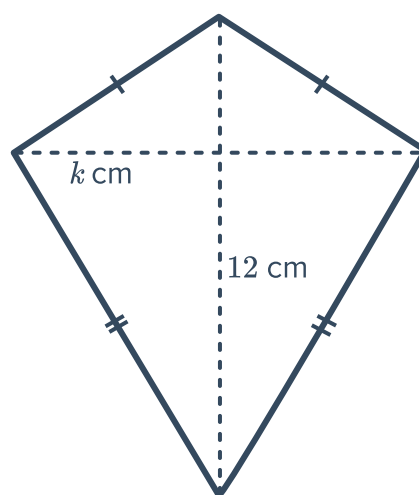


If diagonal $AC = 2$, and diagonal $BD = y$,
find the value of y .



23 The following kite has an area of 48 cm^2 .
The length of one of its diagonals is 12 cm :

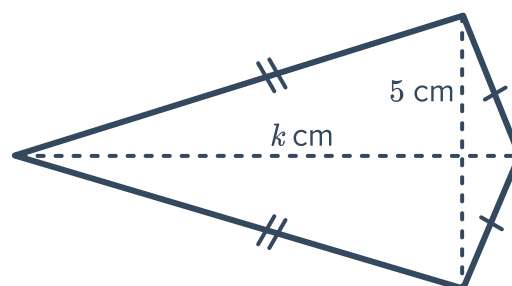
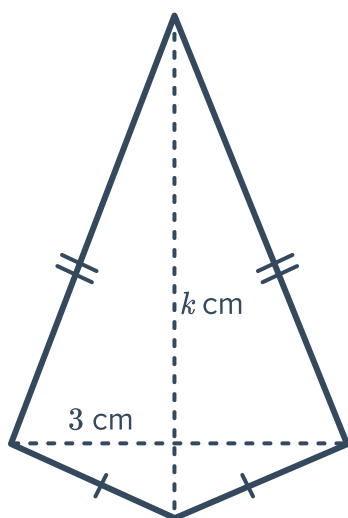
Find the length of the other diagonal, k .



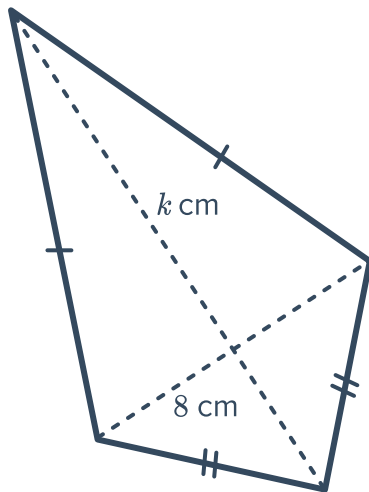
24 For each of the following kites, find the value of k :

a $A = 15 \text{ cm}^2$

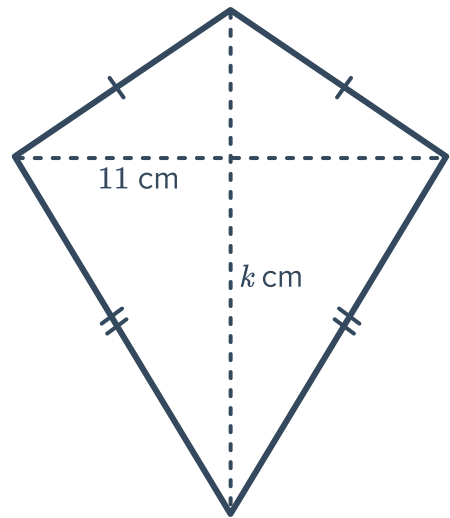
b $A = 22.5 \text{ cm}^2$



c $A = 56 \text{ cm}^2$

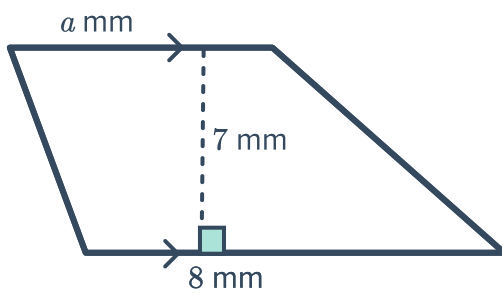


d $A = 137.5 \text{ cm}^2$

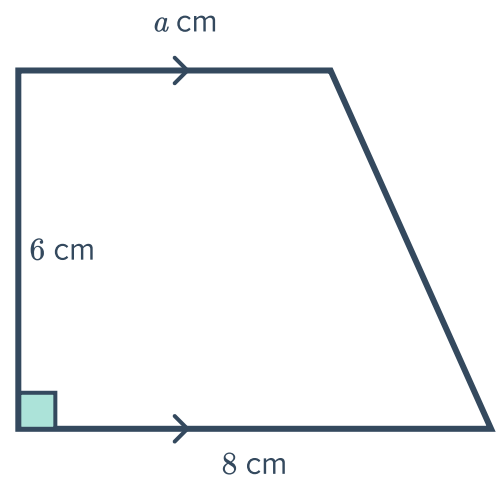


25 For each of the following trapezia, find the value of the pronumeral:

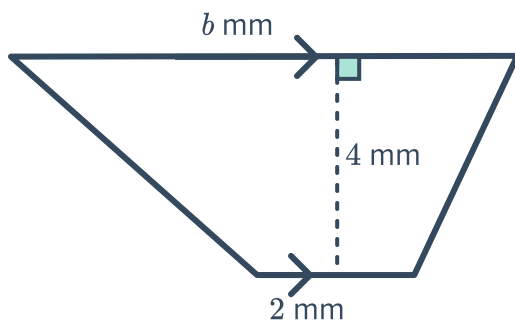
a $A = 42 \text{ mm}^2$



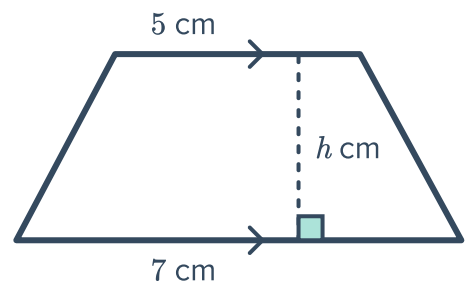
b $A = 36 \text{ cm}^2$



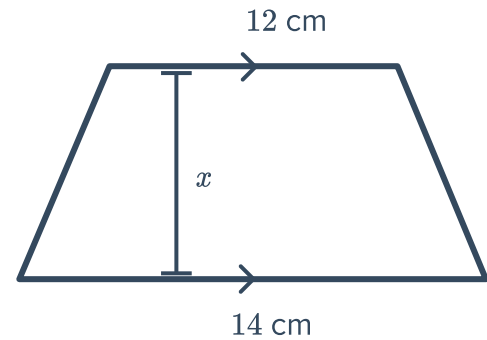
c $A = 20 \text{ m}^2$



d $A = 24 \text{ cm}^2$



- 26 Find the value of x if the area of the trapezium shown is 65 cm^2 :



- 27 Find the perpendicular height, h , of a parallelogram that has an area of 45 cm^2 and a base of 5 cm .
- 28 Find the base length, b , of a parallelogram that has an area of 216 mm^2 and a perpendicular height of 12 mm .
- 29 The area of a kite is 640 cm^2 and one of the diagonals is 59 cm . If the length of the other diagonal is $y \text{ cm}$, find the value of y , rounded to two decimal places.

- 30 Complete the table of base and height measurements for three parallelograms that all have an area of 24 m^2 :

Area (m^2)	Base (m)	Height (m)
24		8
24	12	
24	6	

- 31 Complete the table of the lengths of diagonal x and diagonal y for three kites that all have an area of 36 mm^2 :

Area (mm^2)	Diagonal, x (mm)	Diagonal, y (mm)
36		18
36	24	
36	12	